information, as follows:

```
make mrproper
```

In the kernel configure menu list, choose which features and drivers you need to compile, as follows:

```
make menu config make clean
make depend
```

After configuring the kernel, make a compressed image of the kernel, as shown below:

```
make bzImage
ls -l arch/x86/boot/bzImage
```

Compile the kernel modules, as follows:

```
make modules
```

Runnable kernels are expected to be in the */boot* directory.

```
cp arch/x86/boot/bzImage /boot/vmlinuz-4.2.1
```

To install kernel modules in the */lib/modules/* directory, use the following command:

```
make modules install
```

Set the *initrd* image of the kernel that provides the initial file systems that get loaded into memory during the Linux startup process, as follows:

```
mkinitrd or initramfs /boot/initrd-4.2.1.img 4.2.1
```

After installing the kernel, the GRUB file has to be modified because it contains the details of the kernel's local directory and version:

```
sudo update-grub
```

Power consumption

For power saving in the Linux kernel, TLP can be used. Again, for every kernel job, root privileges are needed. TLP will increase the battery life using the default configuration with the AC and electric battery mode enabled. In AC mode, performance is the most well-liked, whereas in battery mode, power savings is a major concern. TLP accomplishes power saving with the help of dynamic, varied states and timeouts of assorted system devices to scale back the power they draw.

System logging activity

In a default distro installation, complete system logging is regularly arranged. This is suitable for a server or multi-client system. In any case, on a single client system with steady composition, the numerous system log documents will bring about decreasingly intuitive system execution, and lower logging movement will be advantageous for execution and the lifetime of the flash memory. Similar kernel messages are signed in both *kern.log* and *syslog*.

Tools to help you analyse performance

Phoronix Test Suite: The Phoronix Test Suite (PTS) is a free, open source benchmark software for Linux and other operating systems. It is the most comprehensive testing and benchmarking platform available and provides an extensible framework for testing.
Bootchart: Bootchart is a tool for performance analysis and visualisation of GNU/Linux.
Jmeter: This is a 100 per cent pure Java application for load and performance testing. Jmeter is designed to cover categories of tests like load, functional, performance, regression, etc, and requires JDK 5 or higher version.
top: This displays a listing of the most CPU-intensive tasks on the system, and can provide an interactive interface for manipulating processes. It can sort the tasks by CPU usage, memory usage and runtime.
IOzone: IOzone is a file system benchmark tool. The benchmark generates and measures a variety of file operations. It has been ported to many machines and runs under many operating systems.

END

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